

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	pranitha	Reg. No.:	192472073
Section / Batch:	be cse ai	Date:	08.08.26

Code of Conduct

I _____pranitha_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____pranitha_____

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus: *"Data science is powerful, data science drives innovation, data science is evolving."*

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences. The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} | \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

1. MLE of Bigram Probability $P(\text{science} | \text{data})$

Formula:

$$P(\text{science}|\text{data})=\frac{C(\text{data}\ \text{science})}{C(\text{data})}$$

Given:

- $C(\text{data science}) = 3$
- $C(\text{data}) = 3$

$$P(\text{science}|\text{data})=\frac{3}{3}=1.0$$

Answer: 1.0

Interpretation:

The probability is **100%**, meaning whenever the model sees "**data**", it strongly predicts "**science**". In a mobile keyboard, this helps the keyboard quickly suggest **science** after the user types **data**.

2. Backoff Model – "data science improves"

The sequence "**data science improves**" is unseen in the corpus.

- **data science improves** → Trigram unavailable
- **science improves** → Bigram unavailable
- **improves** → Unigram unavailable

Therefore, using the given corpus statistics:

```
[
P(improves|data\ science)=0
]
```

Answer: 0

Why Backoff is essential:

Backoff allows the keyboard to use lower-order models when a higher-order n-gram is unseen. This prevents the system from failing completely on new word combinations and improves real-time prediction.

3. Deleted Interpolation – "data science is"

Formula:

[

P(is|data,science)

```
\lambda_1P(is|data,science)
+\lambda_2P(is|science)
+\lambda_3P(is)
]
```

Given:

- $\lambda_1 = 0.5$
- $\lambda_2 = 0.3$
- $\lambda_3 = 0.2$
- $C(\text{data science is}) = 2$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science}) = 3$
- $C(\text{is}) = 2$
- Total words = 13

Therefore:

$$P(\text{is}|\text{data, science})=\frac{2}{3}=0.667$$

$$P(\text{is}|\text{science})=\frac{2}{3}=0.667$$

$$P(\text{is})=\frac{2}{13}=0.154$$

Now:

$$P=0.5(0.667)+0.3(0.667)+0.2(0.154)$$

$$P=0.3335+0.2001+0.0308$$

$$\boxed{P\approx 0.564}$$

Answer: ≈ 0.564

Interpretation:

Deleted interpolation combines **trigram, bigram, and unigram probabilities**. This provides both **accuracy** from higher-order models and **coverage** when some word sequences are sparse or unseen.

4. Entropy

Given:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Formula:

$$H=-\sum P(x)\log_2 P(x)$$

$$H=-(0.66\log_2 0.66+0.33\log_2 0.33)$$

$$H\approx 0.924$$

$$\boxed{H\approx 0.92 \text{ bits}}$$

Interpretation:

The entropy is approximately **0.92 bits**, indicating relatively low uncertainty because **"is"** has a much higher probability than **"drives"**.

In a smart keyboard:

- **Low entropy** → prediction is more certain.
 - **High entropy** → prediction is more uncertain.
 - Entropy helps the keyboard decide how confident it should be when suggesting the next word.
-

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

1. Penn Treebank POS Tags

Sentence 1: "Book a flight ticket now."

P	Mea
O	nin
S	g
T	
a	
g	
V	Ver
B	b
D	Det
T	ermi
	ner
N	Nou
N	n

N	Nou
N	n

R	Adv
B	erb

Book → **VB** because it is used as an **action** (requesting to book a flight).

Sentence 2: "This book is interesting."

Wo rd	P O S T a g	Mea nin g
Thi s	D T	Det ermi ner
bo ok	N N	Nou n
is	V B Z	Ver b
int ere stin g	J J	Adj ecti ve

book → **NN** because it refers to a **thing/object**, not an action.

2. HMM Probability for "Book" as VB

Given:

[
P(book|VB)=0.6
]

[
 $P(\text{Start} \rightarrow \text{VB}) = 0.5$
]

Using:

[
 $P(\text{VB}, \text{book}) = P(\text{Start} \rightarrow \text{VB}) \times P(\text{book} | \text{VB})$
]

[
 $= 0.5 \times 0.6$
]

[
 $\boxed{= 0.30}$
]

For NN:

[
 $P(\text{NN}, \text{book}) = 0.5 \times 0.4 = 0.20$
]

Therefore:

[
 $\boxed{P(\text{VB}) = 0.30 > P(\text{NN}) = 0.20}$
]

So the HMM favors **VB**.

Reason: The emission probability of *book* given VB (0.6) is higher than its probability given NN (0.4). Also, the sentence context indicates that *Book* is an action.

3. Rule-Based vs HMM Tagging

Rule-Based Tagging

Uses manually written rules

Simple to implement

Less flexible

Difficult with complex sentences

Requires many rules

HMM Tagging

Uses probabilities from training data

Handles ambiguity better

More flexible

Can use contextual information

Learns patterns statistically

Recommendation

HMM/Stochastic tagging is more suitable for a large-scale chatbot because it can handle **ambiguous words based on context**.

For example:

- **Book a flight** → Book = VB
- **This book is interesting** → book = NN

Thus, HMM tagging can provide more accurate POS information for real-world chatbot conversations.

4. Role of POS Tagsets and Word Classes

Standardized POS tagsets such as the **Penn Treebank tagset** help the chatbot understand the grammatical role of words.

Benefits:

1. **Intent Detection** – Identifies actions and objects in user requests.
2. **Response Generation** – Helps generate grammatically correct responses.
3. **Ambiguity Resolution** – Distinguishes words like *book* as VB or NN.
4. **Information Extraction** – Helps identify important nouns, verbs, and entities.
5. **Better NLP Performance** – Provides structured linguistic information for downstream AI models.

Conclusion

POS tagging helps the chatbot understand **what each word is doing in a sentence**. Combining standardized POS tags with HMM-based probabilistic tagging improves **context understanding, intent detection, and response generation**.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
 - economic (120 occurrences)
 - growth (450 occurrences)
 - increases (210 occurrences)
 - employment (380 occurrences)

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

1. Apply the Transformation Rule

Initial tags:

5. economic/JJ growth/NN increases/NNS employment/NN

Rule:

Change NNS → VBZ if the preceding word is tagged NN.

The word **increases** is tagged NNS, and the preceding word **growth** is tagged NN.

Therefore:

6. economic/JJ growth/NN increases/VBZ employment/NN

Corrected POS Sequence

economic/JJ growth/NN increases/VBZ employment/NN

Why?

"Increases" is the main verb of the sentence. The subject is "**growth**", which is singular, so the verb should be **VBZ** (third-person singular present).

2. Initial Tagging Errors

Initial

- 7. economic/JJ
- 8. growth/NN
- 9. increases/NNS
- 10. employment/NN

Corrected

- 11. economic/JJ
- 12. growth/NN
- 13. increases/VBZ
- 14. employment/NN

Word	Initial	Corrected	Reason
economic	JJ	JJ	Describes growth
growth	NN	NN	Subject/noun
increases	NNS	VBZ	Singular present-tense verb
employment	NN	NN	Object/noun

The main error is **increases/NNS**. Here, *increases* is not a plural noun; it is the **verb** describing what economic growth does.

3. Word Frequency Distribution

Given frequencies:

- economic = 120
- growth = 450
- increases = 210
- employment = 380

Total frequency

```
[
120+450+210+380=1160
]
```

Probability of each word

```
[
P(word)=\frac{Frequency(word)}{Total\ Frequency}
]
```

Word	Frequency	Probability
economic	120	0.1034
growth	450	0.3879
increases	210	0.1810
employment	380	0.3276

For example:

```
[
P(growth)=\frac{450}{1160}=0.3879
]
```

Explanation

Frequency information helps probabilistic POS taggers estimate how likely words and tags are. Frequently occurring words provide stronger statistical evidence, helping the system make better tagging decisions when combined with **context and grammatical patterns**.

4. Transformation-Based Tagging and Entropy

How Transformation-Based Tagging improves accuracy

The initial tagger produces:

15. economic/JJ growth/NN increases/NNS employment/NN

The Brill-style transformation identifies the contextual error:

16. NNS → VBZ

because the previous word **growth** is tagged NN.

Final result:

17. economic/JJ growth/NN increases/VBZ employment/NN

Thus, the transformation-based tagger **corrects systematic POS tagging errors using contextual rules**.

Entropy and uncertainty

Entropy measures uncertainty in POS tag assignments.

```
[  
H=-\sum P(tag)\log_2P(tag)  
]
```

- **High entropy** → high uncertainty about the POS tag.
- **Low entropy** → greater confidence in the POS tag.

Before transformation, the word **increases** may have uncertainty between **NNS** and **VBZ**.

After applying the learned rule, **VBZ** becomes the more likely tag, reducing uncertainty.

Conclusion

Transformation-Based Tagging combines **initial statistical tagging with learned linguistic rules**. This improves POS accuracy, especially for ambiguous words, while entropy provides a way to measure how uncertain the system is about its tagging decisions.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____